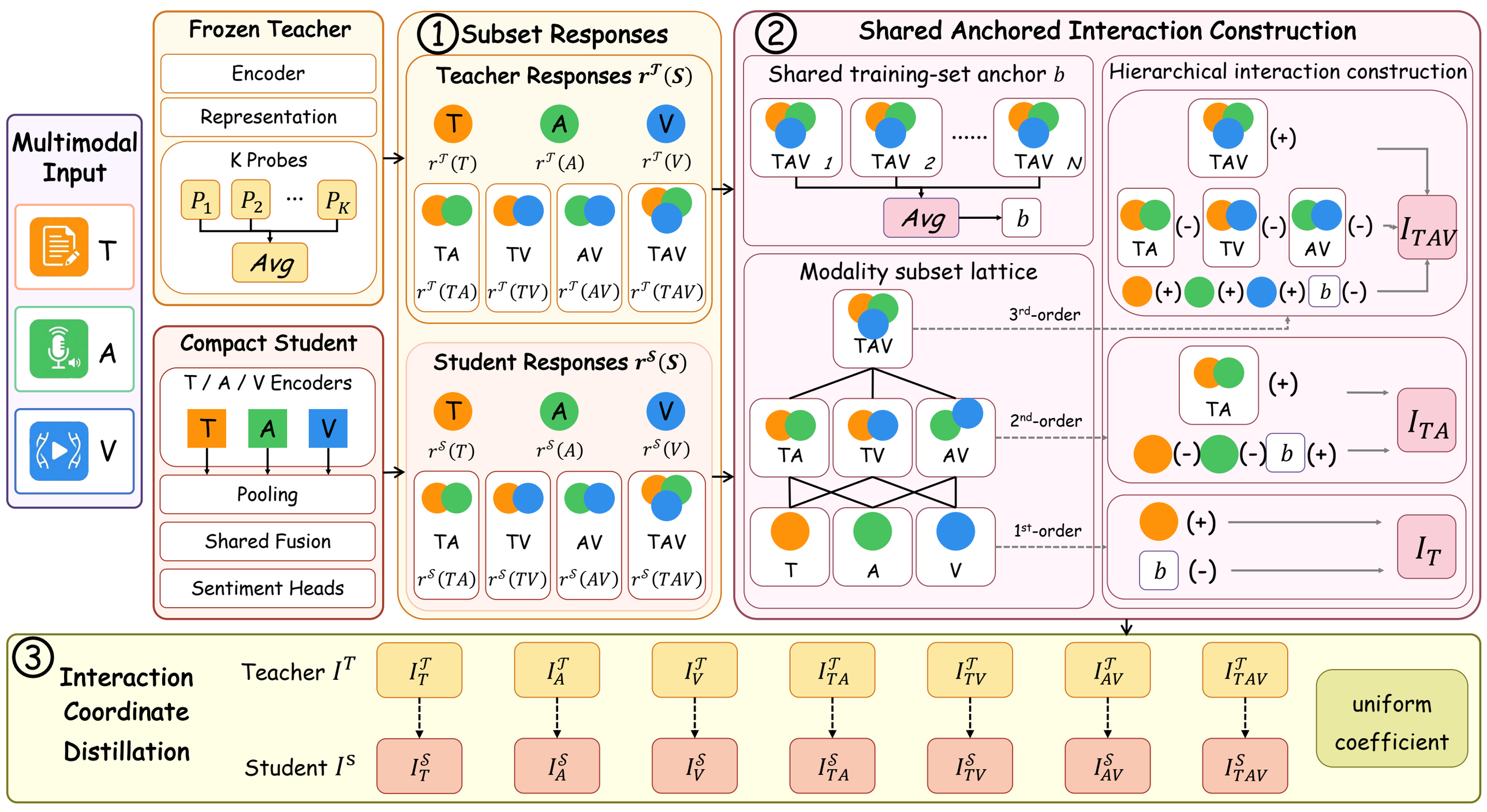

Multimodal Input
T
A
V
Frozen Teacher
Encoder
Representation
K Probes
P_1 P_2 ... P_K
Avg
Compact Student
T / A / V Encoders
T A V
Pooling
Shared Fusion
Sentiment Heads
① Subset Responses
Teacher Responses $r^{\mathcal{T}}(S)$
$r^{\mathcal{T}}(T)$ $r^{\mathcal{T}}(A)$ $r^{\mathcal{T}}(V)$
TA $r^{\mathcal{T}}(TA)$
TV $r^{\mathcal{T}}(TV)$
AV $r^{\mathcal{T}}(AV)$
TAV $r^{\mathcal{T}}(TAV)$
Student Responses $r^{\mathcal{S}}(S)$
$r^{\mathcal{S}}(T)$ $r^{\mathcal{S}}(A)$ $r^{\mathcal{S}}(V)$
TA $r^{\mathcal{S}}(TA)$
TV $r^{\mathcal{S}}(TV)$
AV $r^{\mathcal{S}}(AV)$
TAV $r^{\mathcal{S}}(TAV)$
② Shared Anchored Interaction Construction
Shared training-set anchor b
TAV 1
TAV 2
.....
TAV N
Avg
b
Modality subset lattice
TAV
TA TV AV
T A V
3rd-order
2nd-order
1st-order
Hierarchical interaction construction
TAV (+)
TA (-) TV (-) AV (-) I_{TAV}
(+) (+) (+) b (-)
TA (+)
(-) (-) b (+)
I_{TA}
(+)
b (-)
I_T
③ Interaction Coordinate Distillation
Teacher I^T
Student I^S
$I_T^{\mathcal{T}}$ $I_A^{\mathcal{T}}$ $I_V^{\mathcal{T}}$ $I_{TA}^{\mathcal{T}}$ $I_{TV}^{\mathcal{T}}$ $I_{AV}^{\mathcal{T}}$ $I_{TAV}^{\mathcal{T}}$
$I_T^{\mathcal{S}}$ $I_A^{\mathcal{S}}$ $I_V^{\mathcal{S}}$ $I_{TA}^{\mathcal{S}}$ $I_{TV}^{\mathcal{S}}$ $I_{AV}^{\mathcal{S}}$ $I_{TAV}^{\mathcal{S}}$
uniform coefficient